



MATHEMATICS (EXTENSION 1)

2012 HSC Course Assessment Task 3 (Trial Examination)

June 27, 2012

General instructions

- Working time – 2 hours.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer sheet provided (numbered as page 9)

SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Class (please ✓)

☐ 12M4A – Mr Weiss

☐ 12M3C – Ms Ziazaris

☐ 12M4B – Mr Ireland

☐ 12M3D – Mr Lowe

☐ 12M4C – Mr Fletcher

☐ 12M3E – Mr Lam

Marker's use only.

QUESTION	1-10	11	12	13	14	Total	%
MARKS	$\frac{\quad}{10}$	$\frac{\quad}{15}$	$\frac{\quad}{15}$	$\frac{\quad}{15}$	$\frac{\quad}{15}$	$\frac{\quad}{70}$	

Section I: Objective response

Mark your answers on the multiple choice sheet provided.

Marks

1. Which is the correct value of $\lim_{x \rightarrow 0} \frac{3x}{\sin 2x}$? **1**
 (A) 0 (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) 3

2. Which of the following is the acute angle (correct to the nearest degree) between the two lines $2x - y + 1 = 0$ and $3x + y - 4 = 0$? **1**
 (A) 11° (B) 45° (C) 79° (D) 135°

3. Which of the following expressions will result in the coordinates of the point P which divides the interval AB externally in the ratio $3 : 2$, given A is $(-5, 2)$ and B is $(4, 5)$? **1**
 (A) $\left(\frac{(2)(-5) + (-3)(4)}{-3 + 2}, \frac{(2)(2) + (-3)(5)}{-3 + 2} \right)$
 (B) $\left(\frac{(-3)(-5) + (2)(4)}{-3 + 2}, \frac{(-3)(2) + (2)(5)}{-3 + 2} \right)$
 (C) $\left(\frac{(-2)(-5) + (-3)(4)}{-3 + 2}, \frac{(-2)(2) + (-3)(5)}{-3 + 2} \right)$
 (D) $\left(\frac{(2)(2) + (-3)(5)}{-3 + 2}, \frac{(2)(-5) + (-3)(4)}{-3 + 2} \right)$

4. Which of the following is the derivative of xe^{2x} ? **1**
 (A) $e^{2x}(1 + x)$ (B) $e^{2x}(1 + 2x)$ (C) $2x^2e^{2x}$ (D) $\frac{1}{2}xe^{2x}$

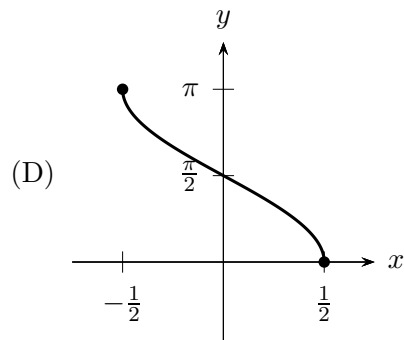
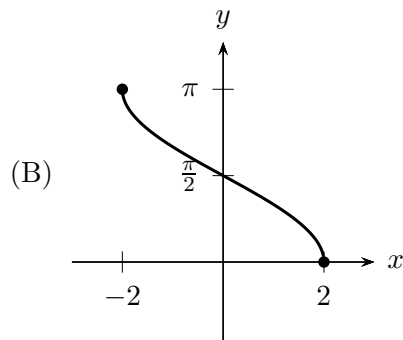
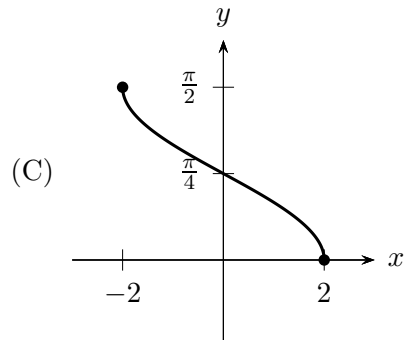
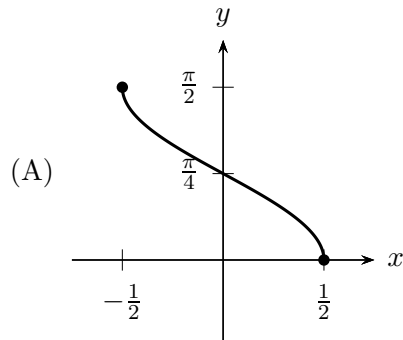
5. Which of the following represents the complete solutions for $-180^\circ < \theta \leq 180^\circ$ to the equation $\cos^2 \frac{\theta}{2} = \frac{1}{4}$ **1**
 (A) $60^\circ, 120^\circ$ (B) $120^\circ, 240^\circ$ (C) $\pm 60^\circ, \pm 120^\circ$ (D) $\pm 120^\circ$

6. What should $\int \cos^2 \frac{1}{2}x \, dx$ be transformed into, in order to find its primitive? **1**
 (A) $\int \frac{1}{2} - \frac{\cos x}{2} \, dx$ (C) $\int \frac{1}{2} - \frac{\cos 2x}{2} \, dx$
 (B) $\int \frac{1}{2} + \frac{\cos 2x}{2} \, dx$ (D) $\int \frac{1}{2} + \frac{\cos x}{2} \, dx$

7. It is known that $\log_e x + \sin x = 0$ has a root close to $x = 0.5$. Using one application of Newton's method, which of the following gives a better approximation to 2 decimal places? 1

(A) 0.43 (B) 0.73 (C) 0.57 (D) 0.27

8. Which of the following graphs represents $y = \cos^{-1} 2x$? 1



9. If $\sqrt{3}\cos x - \sin x \equiv R\cos(x + \alpha)$, which of the following gives the correct value of α ? 1

(A) $\frac{\pi}{6}$ (B) $\frac{5\pi}{6}$ (C) $\frac{7\pi}{6}$ (D) $\frac{11\pi}{6}$

10. Zac and Mitchell play a series of games. The series ends when one player has won two games. In any game the probability that Zac wins is $\frac{3}{5}$ and the probability that Mitchell wins is $\frac{2}{5}$. 1

What is the probability that three games are played?

(A) $\frac{6}{25}$ (B) $\frac{19}{25}$ (C) $\frac{12}{25}$ (D) $\frac{18}{25}$

End of Section I.
Examination continues overleaf.

Section II: Short answer

Question 11 (15 Marks)	Commence a NEW page.	Marks
(a) Solve for x : $\frac{1}{x} > x$		3
(b) Evaluate $\int_0^2 \frac{dx}{\sqrt{16-x^2}}$.		2
(c) Find the exact value of $\sin\left(2\tan^{-1}\frac{3}{7}\right)$, showing full working.		3
(d) Solve $\sin 2\theta = \sin \theta$, $0 \leq \theta \leq 2\pi$.		3
(e) Using the substitution $u = \tan x$, find the exact value of		4

$$\int_0^{\frac{\pi}{4}} \frac{\sec^2 x}{3 + \tan^2 x} dx$$

Question 12 (15 Marks)	Commence a NEW page.	Marks
(a) If $P(x) = x^3 - 6x^2 + ax - 4$, $a > 0$,		
i. Given all the roots of $P(x) = 0$ are real and positive, and that one of the roots is the product of the other two roots, show that $a = 10$.		3
ii. Show that $x - 2$ is a factor of $P(x) = x^3 - 6x^2 + 10x - 4$.		2
(b) Air is being pumped into a spherical balloon at a rate of $20\text{ cm}^3\text{s}^{-1}$. Find the rate of increase of the surface area of the balloon when the radius is 5 cm.		3
(c) A particle moves along the x axis such that its velocity $v\text{ ms}^{-1}$ is given by		

$$v^2 = -4x^2 + 8x + 32$$

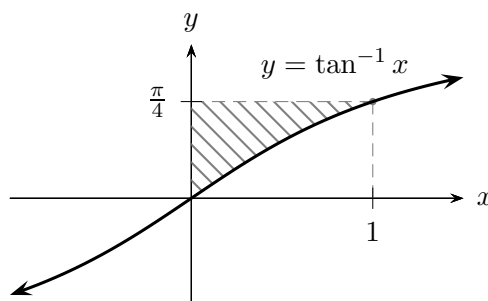
- | | |
|---|---|
| i. By expressing the acceleration as a function in terms of x , prove that the particle is undergoing simple harmonic motion. | 3 |
| ii. Find the amplitude. | 2 |
| iii. Find the maximum acceleration. | 2 |

Question 13 (15 Marks)

Commence a NEW page.

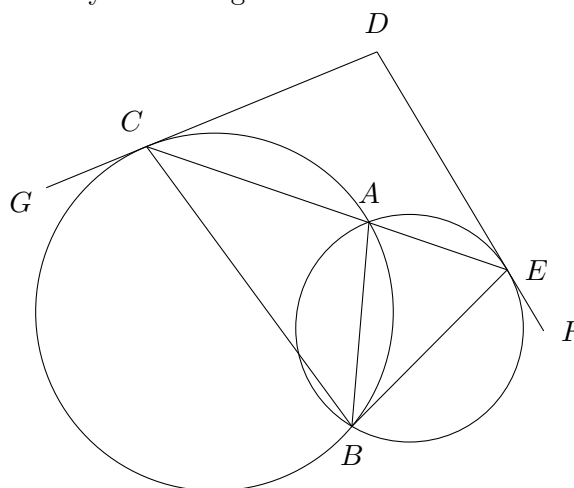
Marks

- (a) Prove by mathematical induction that $5^n + 2 \times 11^n$ is divisible by 3, where n is a positive integer. **3**
- (b) Show that the shaded area is $A = \frac{1}{2} \ln 2$ units². **3**



- (c) Two circles intersect at A and B . CAE is a straight line where C is a point on the first circle and E is a point on the second circle. The tangents to the circles at C and E meet at D . **4**

Copy the diagram into your writing booklet.



Prove that $BCDE$ is a cyclic quadrilateral, without adding any construction lines.

- (d) The acceleration of a raindrop which at time t seconds is falling with speed v metres per second is given by the equation

$$\frac{dv}{dt} = -\frac{1}{3}(v - 3g)$$

where g is a constant.

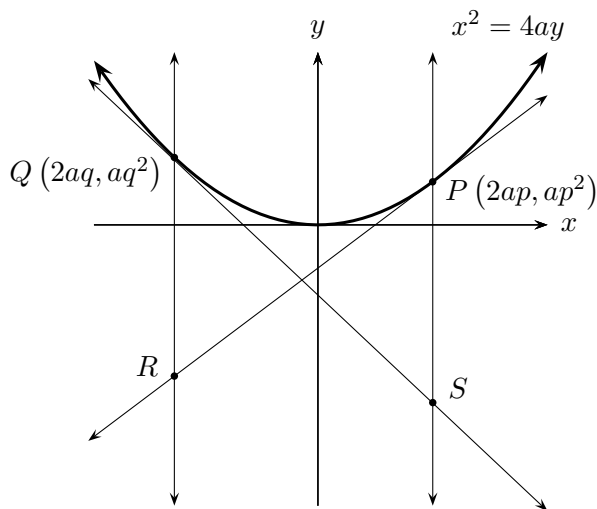
- Show that $v = 3g + Ae^{-\frac{1}{3}t}$, where A is a constant, satisfies the above equation. **1**
- Given that the initial velocity has a value of g , find the value of A . **1**
- After how many seconds is the raindrop falling with a speed of $2g$ metres per second? Give your answer correct to 1 decimal place. **2**
- What value does v approach as $t \rightarrow \infty$? **1**

Question 14 (15 Marks)

Commence a NEW page.

Marks

- (a) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are two points on the parabola $x^2 = 4ay$. The tangent at P and the line through Q parallel to the axis of the parabola meet at the point R .



The tangent at Q and the line through P parallel to the axis of the parabola meet at the point S .

- i. Show that the equations of the tangents at P and Q are $y = px - ap^2$ and $y = qx - aq^2$ respectively. **2**
- ii. Show that the coordinates of S and R are **2**

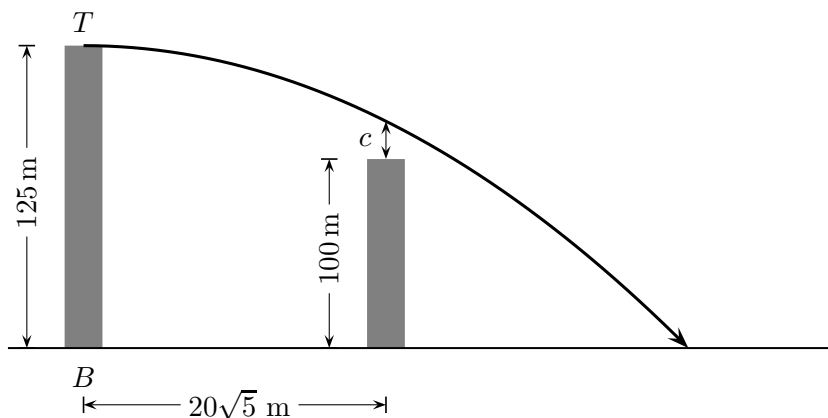
$$S(2ap, 2apq - aq^2) \quad R(2aq, 2apq - ap^2)$$

- iii. Show that $PQRS$ is a parallelogram. **2**
- iv. Show that the area of this parallelogram is $2a^2 |p - q|^3$. **2**

Question 14 continues overleaf...

Question 14 continued from the previous page...

- (b) A projectile is thrown horizontally from the top of a 125 m tower with velocity V metres per second. It clears a second tower of height 100 m by a distance of c metres, as shown. The two towers are $20\sqrt{5}$ metres apart.



- i. The equations of motion for this system are

1

$$\begin{cases} x = Vt \\ y = -5t^2 + 125 \end{cases}$$

(Do not prove this)

Where is the origin of the system being taken from?

- ii. Show that $V = \frac{100}{\sqrt{25 - c}}$.
- iii. Prove that the minimum initial speed of the projectile to just clear the 100 m tower is 20 ms^{-1} .
- iv. Hence, find how far past the 100 m tower will the projectile strike the ground.
- v. Determine the vertical component of the velocity of the projectile when it strikes the ground.

2

1

2

1

End of paper.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g. “●”

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- | | | | | |
|-------------|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 2 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 3 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 4 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 5 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 6 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 7 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 8 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 9 – | ☐ A | ☐ B | ☐ C | ☐ D |
| 10 – | ☐ A | ☐ B | ☐ C | ☐ D |

Suggested Solutions

Section I

(Lowe)

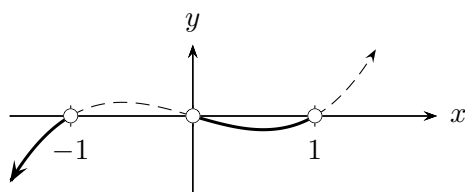
1. (C) 2. (B) 3. (A) 4. (B) 5. (D)
6. (D) 7. (C) 8. (D) 9. (A) 10. (C)

Question 11 (Lam)

(a) (3 marks)

- ✓ [1] for multiplying by square of denominator.
- ✓ [0] for entire part if only multiplying by denominator.
- ✓ [1] for each correct inequality.

$$\begin{aligned}\frac{1}{x} &> \frac{x}{x^2} \\ x &> x^3 \\ x^3 - x &< 0 \\ x(x^2 - 1) &< 0 \\ x(x - 1)(x + 1) &< 0\end{aligned}$$



From the sketch,

$$x < -1 \text{ or } 0 < x < 1$$

(b) (2 marks)

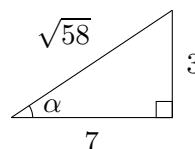
- ✓ [1] for correct primitive.
- ✓ [1] for correct evaluation of limits.

$$\begin{aligned}\int_0^2 \frac{dx}{\sqrt{16-x^2}} &= \left[\sin^{-1} \frac{x}{4} \right]_0^2 \\ &= \sin^{-1} \frac{1}{2} - \sin^{-1} 0 \\ &= \frac{\pi}{6}\end{aligned}$$

(c) (3 marks)

- ✓ [1] for drawing relevant right-angled triangle.
- ✓ [1] for expanding $\sin 2\alpha$.
- ✓ [1] for final answer.

Let $\alpha = \tan^{-1} \frac{3}{7}$. Then $\tan \alpha = \frac{3}{7}$:



$$\begin{aligned}\sin \left(2 \tan^{-1} \frac{3}{7} \right) &\equiv \sin 2\alpha \\ &= 2 \sin \alpha \cos \alpha \\ &= 2 \times \frac{3}{\sqrt{58}} \times \frac{7}{\sqrt{58}} \\ &= \frac{42}{58} = \frac{21}{29}\end{aligned}$$

(d) (3 marks)

- ✓ [1] for factorising expression into $\sin \theta (2 \cos \theta - 1) = 0$.
- ✓ [1] for solutions in positive integral multiples of π .
- ✓ [1] for solutions in multiples of $\frac{\pi}{3}$.

$$\begin{aligned}\sin 2\theta &= \sin \theta \\ 2 \sin \theta \cos \theta - \sin \theta &= 0 \\ \sin \theta (2 \cos \theta - 1) &= 0 \\ \sin \theta = 0 \quad \cos \theta &= \frac{1}{2} \\ \theta = 0, \pi, 2\pi \quad \theta &= \frac{\pi}{3}, \frac{5\pi}{3} \\ \therefore \theta &= 0, \pi, 2\pi, \frac{\pi}{3}, \frac{5\pi}{3}\end{aligned}$$

(e) (4 marks)

- ✓ [1] for changing limits.
- ✓ [1] for making algebraic substitution.
- ✓ [1] for correct primitive.
- ✓ [1] for final answer.

$$\int_0^{\frac{\pi}{4}} \frac{\sec^2 x}{3 + \tan^2 x} dx$$

Letting $u = \tan x$,

$$\begin{aligned} \frac{du}{dx} &= \sec^2 x \\ \therefore du &= \sec^2 x dx \\ x = 0 &\Rightarrow u = 0 \\ x = \frac{\pi}{4} &\Rightarrow u = 1 \\ \int_0^{\frac{\pi}{4}} \frac{\sec^2 x}{3 + \tan^2 x} dx &= \int_{u=0}^{u=1} \frac{\sec^2 x dx}{3 + u^2} \\ &= \int_0^1 \frac{du}{3 + u^2} \\ &= \frac{1}{\sqrt{3}} \left[\tan^{-1} \frac{u}{\sqrt{3}} \right]_0^1 \\ &= \frac{1}{\sqrt{3}} \left(\tan^{-1} \frac{1}{\sqrt{3}} - \tan^{-1} 0 \right) \\ &= \frac{1}{\sqrt{3}} \times \frac{\pi}{6} \\ &= \frac{\pi}{6\sqrt{3}} \left(= \frac{\pi\sqrt{3}}{18} \right) \end{aligned}$$

Question 12 (Low)

(a) i. (3 marks)

- ✓ [1] for $\alpha\beta = 2$.
- ✓ [1] for $\alpha + \beta = 4$.
- ✓ [1] for final answer.

 $P(x) = x^3 - 6x^2 + ax - 4$. Let the roots be α , β and $\alpha\beta$.

- Sum of roots:

$$\alpha + \beta + \alpha\beta = -\frac{b}{a} = 6 \quad (12.1)$$

- Pairs of roots:

$$\alpha\beta + \alpha^2\beta + \beta^2\alpha = \frac{c}{a} = a \quad (12.2)$$

- Product of roots:

$$\begin{aligned} \alpha\beta(\alpha\beta) &= -\frac{d}{a} = 4 \\ \alpha^2\beta^2 &= 4 \\ \therefore \alpha\beta &= 2 \end{aligned} \quad (12.3)$$

as roots are positive.

Substitute (12.3) into (12.1):

$$\begin{aligned} \alpha + \beta + 2 &= 6 \\ \therefore \alpha + \beta &= 4 \end{aligned} \quad (12.4)$$

Substitute (12.4) into (12.2) to find a :

$$\begin{aligned} \alpha\beta + \alpha\beta(\alpha + \beta) &= a \quad (12.5) \\ 2 + 2(4) &= a \\ \therefore a &= 10 \end{aligned}$$

ii. (2 marks)

- ✓ [2] for correct application of factor theorem.

If $x - 2$ is a factor then $P(2) = 0$.

$$\begin{aligned} P(2) &= 2^3 - 6(2^2) + 10(2) - 4 \\ &= 8 - 24 + 20 - 4 = 0 \end{aligned}$$

(b) (3 marks)

- ✓ [1] for $\frac{dr}{dt}$.
- ✓ [1] for $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$.
- ✓ [1] for final answer.

$$\frac{dV}{dt} = 20 = \frac{dV}{dr} \times \frac{dr}{dt}$$

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dr} = \frac{4}{3} \times \pi \times 3r^2 = 4\pi r^2$$

$$\therefore \frac{dV}{dt} = 20 = 4\pi r^2 \Big|_{r=5} \times \frac{dr}{dt}$$

$$= 4\pi \times 25 \times \frac{dr}{dt}$$

$$\therefore \frac{dr}{dt} = \frac{1}{5\pi}$$

Now $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$

$$SA = 4\pi r^2$$

$$\therefore \frac{dA}{dr} = 8\pi r$$

$$\begin{aligned} \frac{dA}{dt} &= 8\pi r \Big|_{r=5} \times \frac{1}{5\pi} \\ &= 8 \text{ cm}^2 \text{ s}^{-1} \end{aligned}$$

(c) i. (3 marks)

- ✓ [1] for using $\frac{d}{dx}(\frac{1}{2}v^2)$ to find acceleration.
- ✓ [1] obtaining $\ddot{x} = -4x + 4$.
- ✓ [1] factorising and noting form $\frac{dv}{dt} = -n^2(x - x_0)$ for SHM.

$$v^2 = -4x^2 + 8x + 32$$

$$\frac{1}{2}v^2 = -2x^2 + 4x + 16$$

$$\begin{aligned} \ddot{x} &= \frac{dv}{dt} = \frac{d}{dx} \left(\frac{1}{2}v^2 \right) \\ &= \frac{d}{dx} (-2x^2 + 4x + 16) \\ &= -4x + 4 = -4(x - 1) \end{aligned}$$

As acceleration is proportion to the opposite direction of displacement, hence the particle is moving in simple harmonic motion with centre at $x = 1$.

ii. (2 marks)

- ✓ [1] for $x = 4, x = -2$.
- ✓ [1] for finding amplitude.

The amplitude occurs when $\dot{x} = 0$.

$$-4x^2 + 8x + 32 = 0$$

$$x^2 - 2x - 8 = 0$$

$$(x - 4)(x + 2) = 0$$

$$\therefore x = 4, -2$$

$$a = \frac{4 + |-2|}{2} = 3$$

As the centre of motion is $x = 1$ and particle's maximum displacement is 4 or -2 , therefore the amplitude is $a = 3$.

iii. (1 mark)

Maximum acceleration occurs at the amplitude, i.e. $x = 4$ or $x = -2$.

$$\begin{aligned} \ddot{x} &= -4(x - 1) \Big|_{x=-2} \\ &= -4(-2 - 1) = 12 \end{aligned}$$

Question 13 (Ziaziaris)

(a) (3 marks)

- ✓ [1] for proving base case.
- ✓ [1] for inductive step.
- ✓ [1] for required proof.

Let $P(n)$ be the statement $5^n + 2 \times 11^n$ is divisible by 3, i.e.

$$5^n + 2 \times 11^n = 3J$$

where $J \in \mathbb{N}$.

- Base case: $P(1)$:

$$5^1 + 2 \times 11 = 5 + 22 = 27$$

which is divisible by 3. Hence $P(1)$ is true.

- Inductive step:
 - Assume $P(k)$ is true for some $k \in \mathbb{N}, k < n$, i.e.

$$5^k + 2 \times 11^k = 3M$$

where $M \in \mathbb{N}$. Alternatively,

$$5^k = 3M - 2 \times 11^k$$

- Examine $P(k + 1)$:

$$\begin{aligned} &5^{k+1} + 2 \times 11^{k+1} \\ &= 5^k 5^1 + 2 \times 11^k 11^1 \\ &= 5 \left(3M - 2 \times 11^k \right) + 2 \times 11^k \\ &= 3 \times 5M - 10 \times 11^k + 2 \times 11^k \\ &= 3 \times 5M - 8 \times 11^k \\ &= 3 \times 5M + 12 \times 11^k \\ &= 3 \underbrace{\left(5M + 4 \times 11^k \right)}_{\in \mathbb{N}} = 3P \end{aligned}$$

where $P \in \mathbb{N}$. Hence $P(k + 1)$ is true.

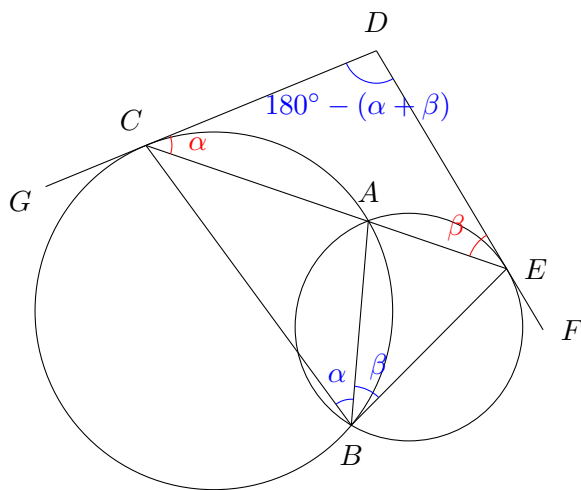
Since $k \in \mathbb{N}$ and truth in $P(k)$ also leads to truth in $P(k + 1)$, therefore $P(n)$ is true by induction.

(b) (3 marks)

- ✓ [1] for converting integrand to $\frac{\sin y}{\cos y}$
- ✓ [1] for correct primitive
- ✓ [1] for final answer.

$$\begin{aligned}
 A &= \int_0^{\frac{\pi}{4}} \tan y \, dy \\
 &= \int_0^{\frac{\pi}{4}} \frac{\sin y}{\cos y} \, dy \\
 &= \left[-\log_e (\cos y) \right]_0^{\frac{\pi}{4}} \\
 &= -\log_e \cos \frac{\pi}{4} + \log_e \cos 0 \\
 &= -\log_e \frac{1}{\sqrt{2}} + \log_e 1 \\
 &= -\log_e 2^{-\frac{1}{2}} = \frac{1}{2} \log_e 2
 \end{aligned}$$

(c) (4 marks) – marking scheme embedded inline. Presence of ✓ indicates 1 mark.



- ✓ Let $\angle DCE = \alpha$ and $\angle DEC = \beta$.
 $\therefore \angle CBA = \alpha$ ($\angle$ in alternate segment)
- ✓ Similarly, $\angle ABE = \beta$ ($\angle$ in alternate segment)
- ✓ Also, $\angle CDE = 180^\circ - (\alpha + \beta)$.
 (Angle sum of $\triangle CDE$)
 - Hence $\angle CDE + \angle CBE = 180^\circ$
- ✓ Opposite $\angle$ in $BCDE$ are supplementary. Hence $BCDE$ is a cyclic quadrilateral.

(d) i. (1 mark)

$$\begin{aligned}
 \frac{v}{-3g} &= \frac{3g}{-3g} + Ae^{-\frac{1}{3}t} \\
 v - 3g &= Ae^{-\frac{1}{3}t} \\
 \frac{dv}{dt} &= -\frac{1}{3} \underbrace{Ae^{-\frac{1}{3}t}}_{=(v-3g)} \\
 &= -\frac{1}{3}(v - 3g)
 \end{aligned}$$

ii. (1 mark)

$$\begin{aligned}
 t = 0, v &= g \\
 \therefore g &= 3g + Ae^0 \\
 \therefore A &= -2g
 \end{aligned}$$

iii. (2 marks)

$$\begin{aligned}
 v &= 2g, t = ? \\
 2g &= 3g - 2ge^{-\frac{1}{3}t} \\
 -g &= -2ge^{-\frac{1}{3}t} \\
 \frac{1}{2} &= e^{-\frac{1}{3}t} \\
 -\frac{1}{3}t &= \log_e \frac{1}{2} = -\log_e 2 \\
 \therefore t &= 3 \log_e 2 \approx 2.1 \text{ seconds}
 \end{aligned}$$

iv. (1 mark)

As $t \rightarrow \infty$, $v \rightarrow 3g$.

Question 14 (Ireland/Fletcher)

(a) i. (2 marks)

✓ [1] for proving $\frac{dy}{dx} = p$ at P .✓ [1] for equation of tangent at P .

$$x^2 = 4ay \Rightarrow y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a} = \frac{x}{2a}$$

At $x = 2ap$,

$$\frac{dy}{dx} = \frac{2ap}{2a} = p$$

Equation of the tangent at P :

$$y - ap^2 = p(x - 2ap) = px - 2ap^2$$

$$y = px - ap^2$$

Similarly, the tangent at Q is

$$y = qx - aq^2$$

ii. (2 marks)

✓ [1] for each y coordinate.Coordinates of S arise from the intersection of $x = 2ap$ and $y = qx - aq^2$:

$$y = q(2ap) - aq^2 = 2apq - aq^2$$

$$\therefore S(2ap, 2apq - aq^2)$$

Coordinates of R arise from the intersection of $x = 2aq$ and $y = px - ap^2$:

$$y = p(2aq) - ap^2 = 2apq - ap^2$$

$$\therefore R(2aq, 2apq - ap^2)$$

iii. (2 marks)

✓ [1] for showing $PS \parallel QR$.✓ [1] for showing $PS = QR$.

$$d_{PS} = \sqrt{(\cancel{2ap} - 2ap)^2 + (ap^2 - (2apq - aq^2))^2}$$

$$= \sqrt{(a(p - q)^2)^2}$$

$$= a(p - q)$$

$$d_{QR} = \sqrt{(\cancel{2aq} - 2aq)^2 + (aq^2 - (2apq - ap^2))^2}$$

$$= \sqrt{a(q - p)^2} = \sqrt{a(p - q)^2}$$

$$= a(p - q)$$

As $PS = QR$, hence one pair of opposite sides equal and parallel. Hence $PQRS$ is a parallelogram.**Alternatively**, if $PQRS$ is a parallelogram, then the diagonals bisect each other; i.e. QS and PR share the same midpoint. Show via midpoint formula results in

$$MP_{QS} = \left(\frac{a(p + q)}{2}, apq \right)$$

$$MP_{PR} = \left(\frac{a(p + q)}{2}, apq \right)$$

iv. (2 marks)

✓ [1] for h (fully)

✓ [1] for area.

- Use $A = bh$.
- h is perpendicular distance from Q to PS .
- Use $b = d_{PS}$.

Using the perpendicular dist formula with $x = 2ap$ & $Q(2aq, aq^2)$:

$$h = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|2aq(1) + 0 - 2ap|}{\sqrt{1^2 + 0}}$$

$$= \frac{|2aq - 2ap|}{1} = 2a|q - p|$$

$$= 2a|p - q|$$

As $|p - q| = |q - p|$.

$$A = bh$$

$$= 2a|p - q| \times a(p - q)^2$$

$$= 2a^2|p - q|^3$$

- (b) i. (1 mark)
Origin is at the base of tower.

- ii. (2 marks)

✓ [1] for $100 + c = -5 \left(\frac{400 \times 5}{V^2} \right) + 125$.

✓ [1] for final result shown.

When $x = 20\sqrt{5}$, $y = 100 + c$. Using $x = Vt$,

$$20\sqrt{5} = Vt$$

$$\therefore t = \frac{20\sqrt{5}}{V}$$

Substitute into $y = -5t^2 + 125$,

$$100 + c = -5 \left(\frac{20\sqrt{5}}{V} \right)^2 + 125$$

$$= -5 \left(\frac{400 \times 5}{V^2} \right) + 125$$

$$-25 + c = -5 \times \frac{400 \times 5}{V^2}$$

$$25 - c = \frac{25 \times 400}{V^2}$$

$$V^2 = \frac{10\,000}{25 - c}$$

$$\therefore V = \frac{100}{\sqrt{25 - c}} \quad (V > 0)$$

- iii. (1 mark) Projectile just clears tower when $c = 0$.

$$V = \frac{100}{\sqrt{25 - c}} \Big|_{c=0} = 20 \text{ ms}^{-1}$$

- iv. (2 marks)

✓ [1] for $x = 100$.

✓ [1] for final answer.

Projectile strikes ground when $y = 0$.

$$-5t^2 + 125 = 0$$

$$5t^2 = 125$$

$$\therefore t^2 = 25 \Rightarrow t = 5$$

When $t = 5$,

$$x = Vt = 20 \times 5 = 100$$

Hence projectile will strike the ground $100 - 20\sqrt{5}$ metres past the second tower.

- v. (1 mark)

$$y = -5t^2 + 125$$

$$\dot{y} = -10t \Big|_{t=5} = -50 \text{ ms}^{-1}$$